PALESTINE POLYTECHNIC UNIVERSITY
COLLEGE OF INFORMATION TECHNOLOGY AND COMPUTER ENGINEERING
SYSTEMS PROGRAMMING
COURSE PROJECT –SPRING2025

<table>
<tr><td>PROJECT NAME</td><td>SicAssember V18.00</td></tr>
<tr><td>PROJECT OBJECTIVES</td><td>

Implementing an assembler for the SIC hypothetical machine
Improving programming skills

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<tr><td>PROJECT DESCRIPTION</td><td>
The project is a simulation to the assembler of the SIC hypothetical machine. Students should implement both Pass1 and Pass2 of the SIC assembler language. Your assembler should consider all the issues:

Directives: START, END, BYTE, WORD, RESB, RESW.
Comments: If a source line contains a period (.) in the first byte, the entire line is treated as a comment.
Addressing modes: Simple and Indexed.
Instruction Set: Specified in Appendix A of the textbook.
<u>Errors: Your Assembler should flag ALL expected errors</u>

You should include test files. Assume a fixed format source code with all text written in uppercase. The output of Pass 1 is:

Symbol Table SYBTAB: should be displayed on the screen.
LOCCTR, PRGLTH, PRGNAME, …
Intermediate file (.mdt): Stored on the secondary storage.

The output of Pass 2:

The object file (.obj)
The listing file (.lst)
List of errors if happened (duplicate labels, invalid mnemonic, inappropriate operand…).

When your source code has a <u>fixed format</u>, you are committed to the following dimensions:
<pre>
Columns: 1-10 Label
          11-11 Blank
          12-20 Operation code (or Assembler directive)
          21-21 Blank
          22-39 Operand
          40-70 Comment
</pre>
If your project does not have a GUI, then you should specify all the arguments well to the command prompt as follow (Assuming executable version stored on the C partition):
<pre>
C:\> pass1 cource_file.asm intmdte_file.mdt

C:\> pass2 intmdte_file.mdt output_file.obj
</pre>
Your assembler can stop execution if there are errors in Pass 1, and should indicate its termination with appropriate messages.
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PROJECT REGULATIONS	• **You should use PYTHON programming language to implement the project.**
	• The project should be submitted in two formats:
	    ▪ *Soft copy*: Uploaded to Google Class Room.
	    ▪ *Hard copy*: A neat printed document containing the source code.
	• There will be 5% penalty for each late day in submission.
	• **Copied projects will be treated severely.**

PROJECT TEAM	NO project Teams!
SUBMISSION DATE	*PASS I:* Monday, APRIL 07, 2025 *PASS II:* Wednesday, April 23, 2025
DISCUSSION DATE	Each student should contact me when submitting each pass in order to manage the discussion time.

INSTRUCTOR:
YOUSEF A. SALAH